TTT
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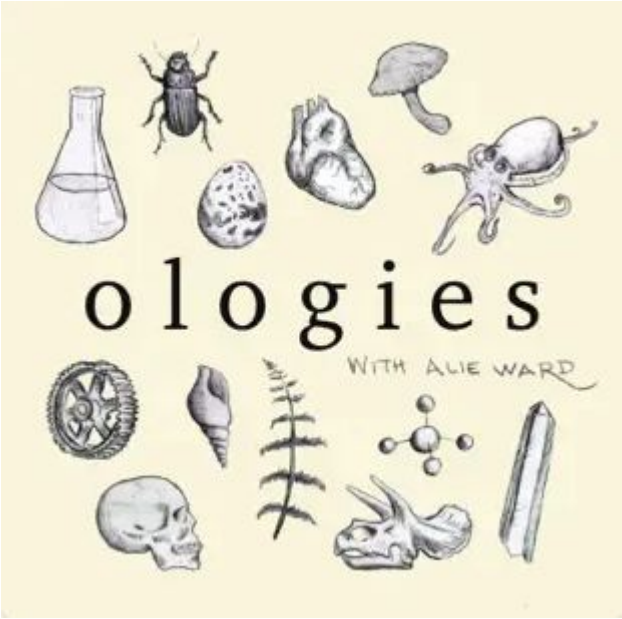
Lauren Sharwood

Podcast Corner - Ologies

https://www.alieward.com/ologies

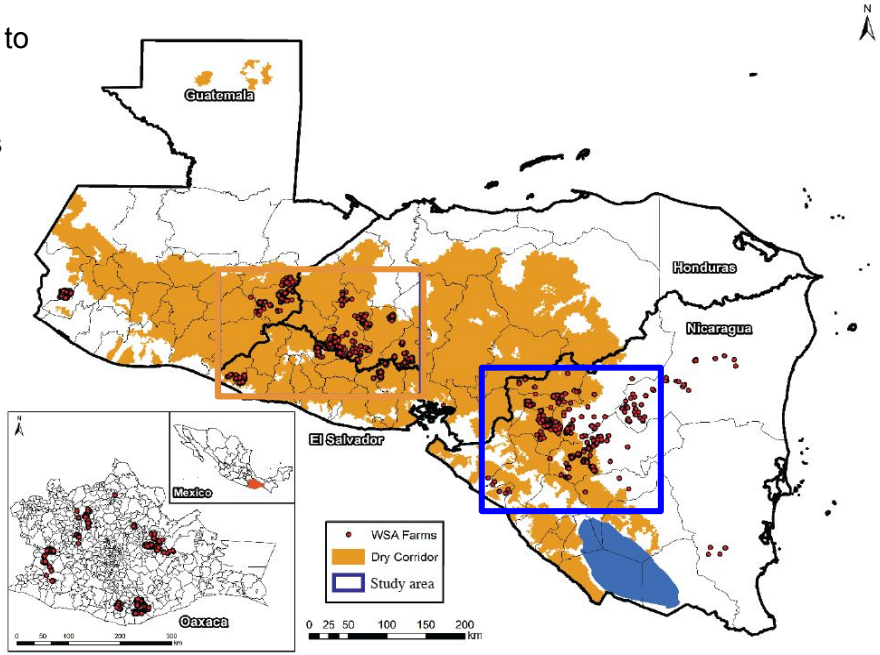


- Science/comedy podcast
- “Alie Ward asks smart people stupid questions”
- Some episodes: dendrology (trees), microbiology (gut microbiome), critical ecology (social systems & environment), zymology (beer), lutrinology (otters)

Catholic Relief Services (CRS) Project Wrap-Up

Context: Climate change has changed rain patterns across Central America's dry corridor, challenging conventional agriculture's capacity to deliver a reliable stream of income for small landholders. CRS is working to address these challenges through the Water Smart Agriculture (WSA) program, a suite of sustainable agricultural practices that seek to increase agricultural resilience to a changing climate and restore degraded agricultural soils. To assess the impacts of CRS's intervention at the landscape scale, there is need for a scalable, affordable method to monitor the adoption of WSA practices across large regions.

Main Objective: To assess the potential of satellite imagery to detect the adoption of two target WaterSmart Agriculture (WSA) practices: crop residue retention and the use of cover crops



Data & Methods

Practices field data:

										Cultivo
		Temporada	Postrera	Primera	Postrera	Primera	Postrera	Primera	Postrera	Primera
		Ano	2016	2016	2017	2017	2018	2018	2019	2019
ID_Prod	Parcela	Nom.Cob								
NI_313	ASA	maiz-frijol-canavalia	Sorgo	Maiz	Frijol	Maiz	NaN	Maiz	Sorgo	Maiz
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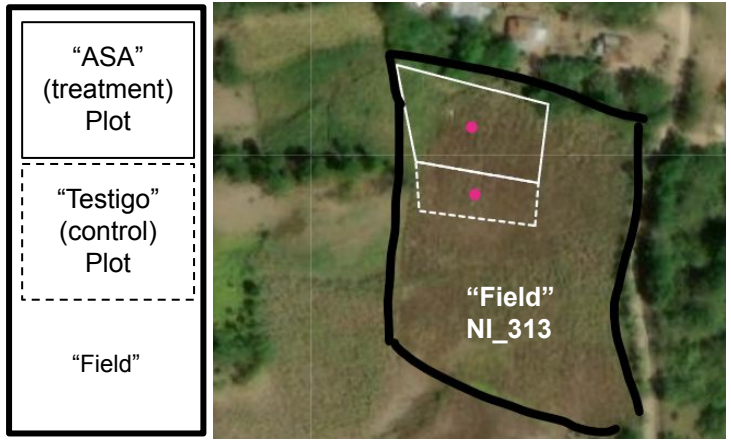
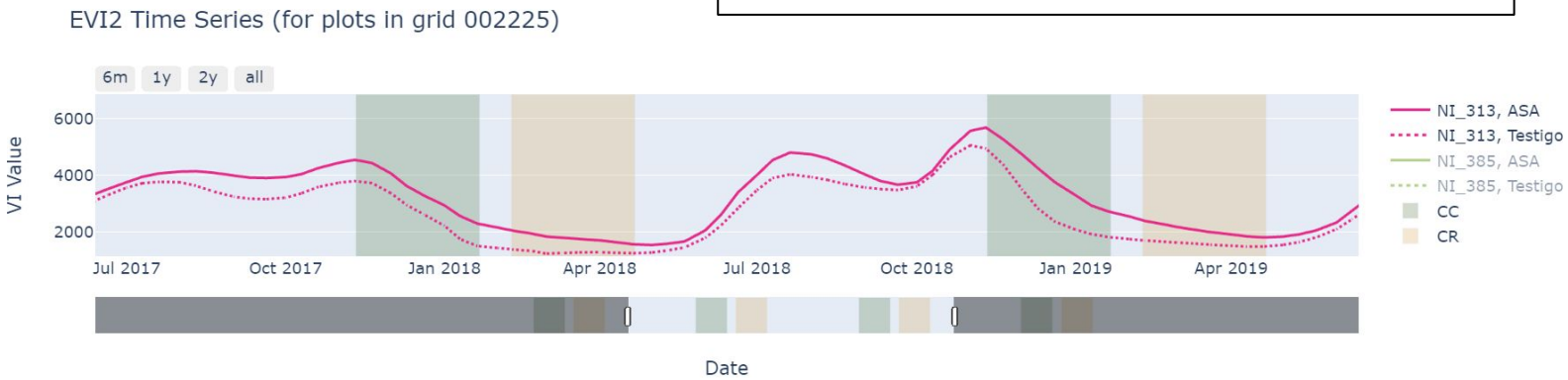


Methods: Download and correct Sentinel-2 and Landsat imagery from 2015 to 2021. Create VIs targeting cover crops (CC) and crop residue (CR) for every image and apply temporal smoothing to the time-series

Jargon:
Vegetation Index (VI): A single number related to vegetation properties (often vigor or biomass) calculated using two or more image bands (certain segments of light, i.e. red, green, blue, NIR, SWIR) that are sensitive to vegetation lignin, moisture, chlorophyll content.

Expect ASA plots to have higher VI (EVI2, GCVI, NBR, WI, NDTI) values during CC/CR windows



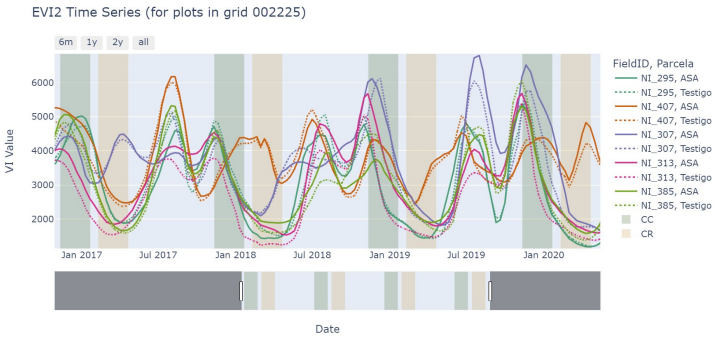
Findings

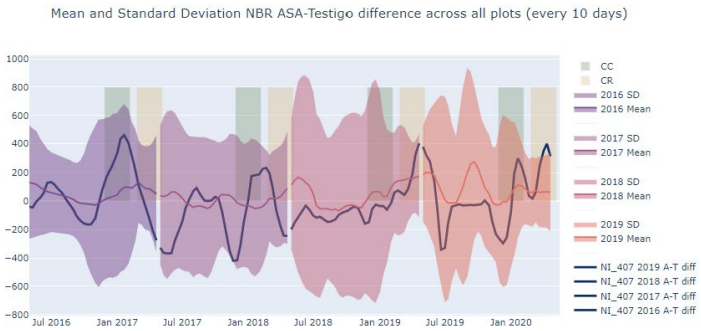
VI Time series:

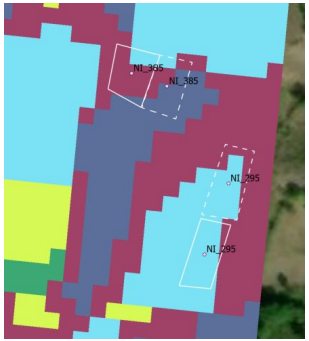
- ASA & Testigo VI time series for a given field more similar to each other than a nearby field's respective plot
- ASA & Testigo VI difference minimal and flipped from year to year

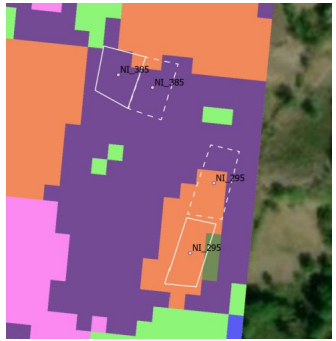
K-means clustering results:

- Some plots clustered separately
- Different input features (VI, VI month, yearly VI min/max/amplitude) best separate ASA & Test plots on different fields





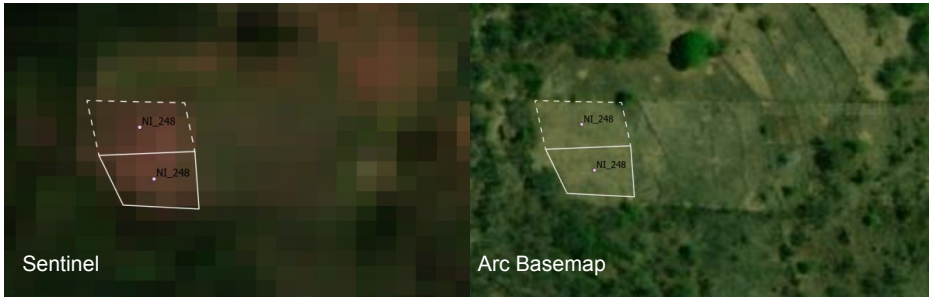




Challenges

Field & Practice data:

- Received complete field boundaries and practice data for 18 Nicaragua pairs
- ASA and Test plot size or ratio
- Test plots began adopting ASA practices, but aren't listed in practice data



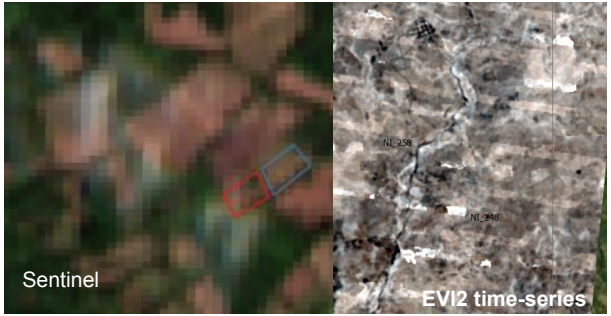
Sentinel-2 & Landsat imagery:

- Sensor striping, clouds, and shadow artifacts in images
- **Pixel size -> Mixed pixels on smallholder fields using 10m & 30m resolution imagery**

Lessons Learned

Improvements to CEL crop-mapping workflow:

- better cloud and shadow masking
- add phenology metrics

Scaling the project would require …

- representative samples to account for all main crop types and crop cycles over the dry corridor
- precise plot location boundaries and practice data with implementation dates
- higher-resolution imagery (that's still affordable over large regions) to automatically segment individual plots